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Z5008: Big Data Laboratory

**Project Proposal: ZECO Revenue Protection & Customer Intelligence Platform:
Real-Time Analytics for Prepaid Electricity**

Domain: Prepaid Electricity Consumption (ZECO Data)

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1 Introduction

The ZECO Revenue Protection and Customer Intelligence Platform is a Big Data analytics system designed to improve prepaid electricity operations through real-time analytics, customer intelligence, forecasting, and revenue protection mechanisms.

The project utilizes historical electricity vending transactions from the Zanzibar Electricity Corporation (ZECO) and transforms them into a modern analytics platform based on Lakehouse architecture, Spark processing, streaming analytics, forecasting models, and revenue protection dashboards.

The primary objective is to enable operational teams to monitor electricity sales, understand customer behaviour, forecast future demand, and identify suspicious activities that may lead to revenue losses.

2 Project Objectives

The objectives of the project are:

1. Build a scalable Lakehouse architecture for ZECO data.
2. Develop real-time streaming analytics using historical replay.
3. Implement customer intelligence dashboards.
4. Detect suspicious customer behaviour for revenue protection.
5. Forecast future electricity demand and revenue.
6. Demonstrate an end-to-end Big Data and Analytics platform.

3 Dataset Description

The project utilizes historical prepaid electricity vending transactions obtained from Zanzibar Electricity Corporation (ZECO).

The dataset contains:

- Meter Number
- Account Number
- Transaction Date
- Purchase Amount (TZS)
- Electricity Units Purchased
- Town
- Customer Activity Information

The platform currently processes approximately 28.8 million transaction records.

3.1 Dataset Purpose

The dataset supports:

- Revenue Analytics
- Customer Intelligence
- Consumption Analysis
- Revenue Protection
- Forecasting
- Real-Time Monitoring

4 System Architecture

The proposed architecture consists of eight layers.

1. Streaming Ingestion Layer
2. Lakehouse Storage Layer
3. Processing Layer
4. Orchestration Layer
5. Machine Learning Layer
6. Experiment Tracking Layer
7. Serving Layer
8. Visualization and Monitoring Layer

4.1 System Architecture

The ZECO Revenue Protection and Customer Intelligence Platform follows a layered architecture consisting of eight interconnected components designed to support scalable data ingestion, storage, analytics, machine learning, forecasting, and decision support. Figure 1 illustrates the overall architecture.

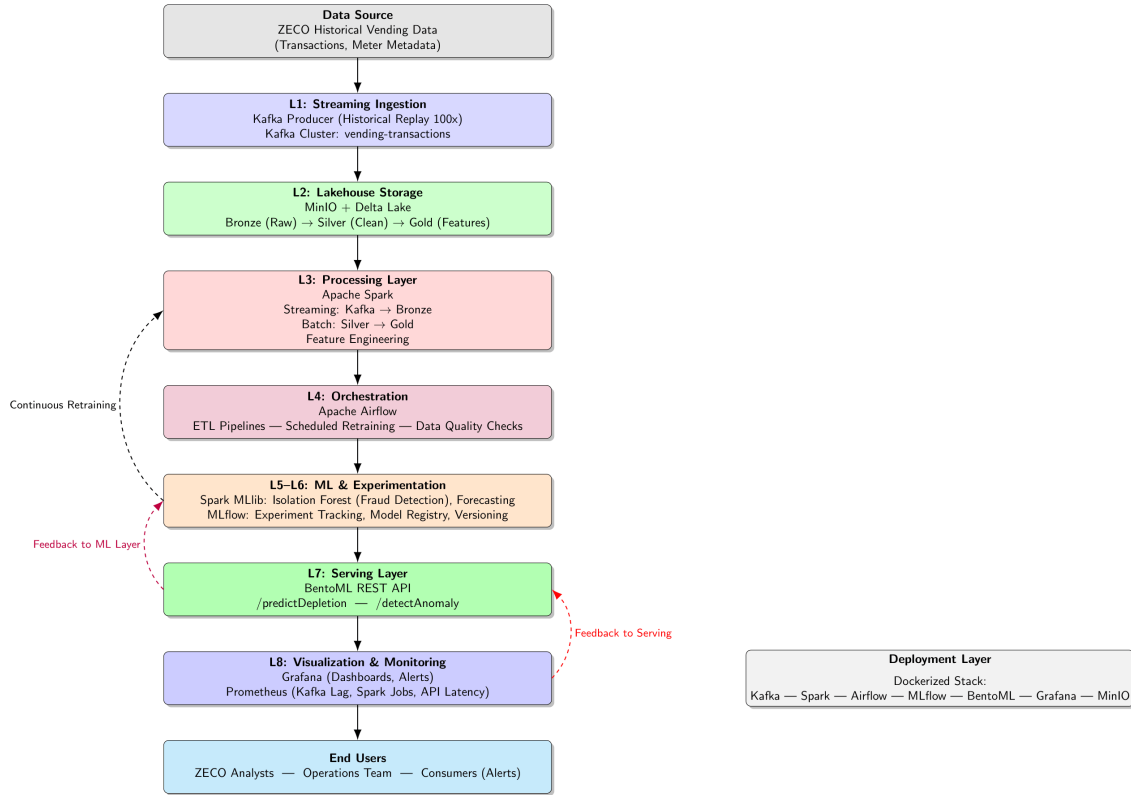


Figure 1: System Architecture of the ZECO Revenue Protection and Customer Intelligence Platform

4.1.1 Streaming Ingestion Layer

The Streaming Ingestion Layer is responsible for collecting and transporting electricity vending transactions into the analytics platform.

Since ZECO data was provided as historical vending records rather than a live stream, Apache Kafka was used to replay historical transactions and simulate a real-time streaming environment. A Kafka producer continuously publishes transaction events to Kafka topics while Kafka consumers receive these events for downstream processing.

Each transaction record contains customer information, meter identifiers, purchased electricity units, transaction amounts, timestamps, and geographical information. This layer provides a scalable foundation for future deployment with real ZECO operational systems.

The main objective of this layer is to ensure continuous event ingestion while decoupling data producers from downstream analytics services.

4.1.2 Lakehouse Storage Layer

The Lakehouse Storage Layer provides centralized storage for all datasets generated throughout the analytics pipeline.

MinIO object storage was deployed as the primary storage engine and organized according to the Medallion Architecture consisting of:

- Bronze Layer: Raw Kafka transaction records.
- Silver Layer: Cleaned and validated datasets.
- Gold Layer: Curated analytical datasets used for dashboards and machine learning.

Apache Spark stores datasets in Delta Lake format, enabling ACID transactions, schema evolution, versioning, and efficient analytical queries.

This architecture separates raw data from processed data and improves maintainability while supporting both batch and streaming workloads.

4.1.3 Processing Layer

The Processing Layer was implemented using Apache Spark and serves as the computational engine of the platform.

Two major processing workflows were developed:

1. Streaming Processing
2. Batch Processing

For streaming workloads, Spark Structured Streaming consumes Kafka messages and writes processed records into Bronze and Silver layers.

For batch workloads, Spark performs:

- Data cleaning
- Missing value handling
- Feature engineering
- Revenue aggregation
- Customer spending calculations
- Town-level electricity consumption analysis
- Fraud indicator generation

Several derived features were generated, including:

$$AveragePurchaseAmount = \frac{\text{Total Amount Spent}}{\text{Transaction Count}} \quad (1)$$

$$ConsumptionRate = \frac{\text{Electricity Units}}{\text{Days Between Purchases}} \quad (2)$$

These engineered features form the basis for forecasting and anomaly detection models.

4.1.4 Orchestration Layer

Apache Airflow was integrated to automate workflow execution and manage dependencies between different pipeline stages.

Airflow Directed Acyclic Graphs (DAGs) were designed to schedule:

- Data ingestion jobs
- Spark ETL pipelines
- Feature generation tasks
- Forecasting model retraining
- Data quality validation checks

This orchestration layer eliminates manual intervention and ensures reproducible execution of the analytics pipeline.

4.1.5 Machine Learning Layer

The Machine Learning Layer provides predictive analytics and revenue protection capabilities.

Two major categories of models were implemented.

Revenue Forecasting Time-series forecasting models were trained using historical vending data.

The AutoRegressive Integrated Moving Average (ARIMA) model was used to forecast future revenue trends:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \epsilon_t + \sum_{j=1}^q \theta_j \epsilon_{t-j} \quad (3)$$

The Prophet forecasting model, originally developed by Facebook Research (now Meta), was implemented to forecast future electricity consumption demand and seasonal usage patterns. Prophet was selected because it effectively models long-term trends, recurring seasonal behaviour, and irregular fluctuations commonly observed in electricity consumption data.

Revenue Protection and Anomaly Detection Isolation Forest was implemented to identify suspicious customer behaviour using unsupervised machine learning. The model was selected because confirmed fraud labels are not available, making unsupervised anomaly detection suitable for this project.

The model was trained using customer-level behavioural features including transaction count, total spending, average purchase value, maximum purchase value, electricity units purchased, purchase variability, customer lifetime, inactive days, depletion interval, purchase spike ratio, unit spike ratio, and consumption drop ratio.

Before training, numerical features were standardized using:

$$z = \frac{x - \mu}{\sigma}$$

where x is the original feature value, μ is the mean, and σ is the standard deviation.

The Isolation Forest model assigns anomaly scores by isolating unusual customer behaviour patterns. Customers with higher anomaly risk scores are more likely to represent suspicious vending behaviour.

The risk score was converted into dashboard categories as follows:

- 0 = Normal
- 1 = Medium Risk
- 2 = High Risk

In addition to the machine learning anomaly score, explainable fraud indicators were also retained to support revenue protection interpretation. These include sudden consumption drops, purchase spikes, irregular vending intervals, inactive customer behaviour, unit consumption anomalies, and high variability behaviour.

The final fraud scenario score is calculated as:

$$FraudScenarioScore = \sum_{k=1}^8 Flag_k$$

where $Flag_k$ represents each activated fraud indicator.

4.1.6 Experiment Tracking Layer

MLflow was integrated to support experiment tracking and model lifecycle management.

The platform records:

- Model parameters
- Training metrics
- Forecasting accuracy
- Model versions
- Execution timestamps

This layer enables reproducibility and systematic comparison of forecasting and anomaly detection models.

4.1.7 Serving Layer

The Serving Layer exposes analytical outputs and machine learning predictions to downstream applications.

BentoML was selected for model deployment and API serving.

REST endpoints were designed to support:

- Revenue forecasting requests
- Consumption forecasting requests

- Fraud risk scoring
- Customer anomaly detection

This layer enables future integration with ZECO operational systems and mobile applications.

4.1.8 Visualization and Monitoring Layer

The Visualization and Monitoring Layer provides decision support for ZECO analysts and management.

A Streamlit-based dashboard was developed containing multiple analytical modules:

- KPI Overview
- Live Streaming Analytics
- Monthly Revenue Analytics
- Town Electricity Usage Analytics
- Customer Spending Analysis
- Revenue Protection Dashboard
- Suspicious Customer Explorer
- Revenue Forecasting Dashboard
- Token Depletion Forecast Dashboard

The dashboard currently reports:

- Total revenue exceeding TZS 221 billion
- More than 28 million transactions analysed
- Electricity units sold across all regions
- Revenue protection alerts
- High-risk customer identification
- Future revenue forecasts
- Token consumption forecasts

Grafana and Prometheus were additionally integrated for infrastructure monitoring, system health tracking, and operational observability.

The architecture follows a modern data engineering pipeline where data flows from raw ingestion through analytical processing and finally into operational dashboards.

5 Implementation Progress

5.1 Lakehouse Architecture

A complete Bronze-Silver-Gold Lakehouse architecture has been implemented.

5.1.1 Bronze Layer

The Bronze layer stores raw ZECO vending transactions exactly as received.

Main objectives:

- Preserve original data
- Enable data lineage
- Support future reprocessing

5.1.2 Silver Layer

The Silver layer performs:

- Data cleaning
- Null value handling
- Schema standardization
- Data validation

5.1.3 Gold Layer

The Gold layer contains business-ready datasets for:

- Revenue analytics

- Customer intelligence
- Town-level analytics
- Fraud detection
- Forecasting

5.2 Spark ETL Pipeline

Apache Spark was used to process ZECO transactions.

The ETL pipeline performs:

1. Data ingestion
2. Data transformation
3. Feature engineering
4. Aggregation
5. Export to Gold datasets

The Spark implementation enables distributed processing of millions of electricity vending records.

6 Real-Time Streaming Analytics

A Spark Structured Streaming pipeline has been implemented.

Historical ZECO transactions are replayed as streaming events to simulate a real production environment.

The streaming system continuously generates:

- Live Revenue
- Live Units Sold
- Live Transactions
- Revenue Protection Alerts

Live revenue is computed as:

$$[\text{Revenue}_t = \sum_{i=1}^n \text{PurchaseAmount}_i]$$

where:

- n = number of transactions in a streaming batch

Live electricity units sold are computed as:

$$[\text{Units}_t = \text{sum}_{i=1}^n \text{ElectricityUnits}_i]$$

The dashboard updates automatically as new streaming batches arrive.

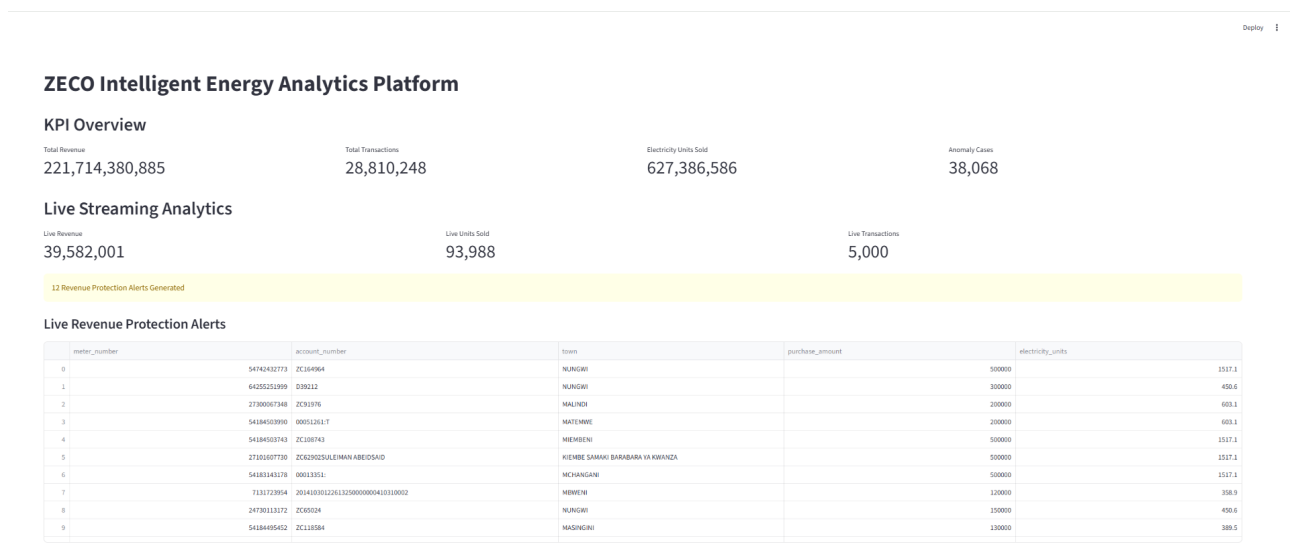


Figure 2: Live Streaming Analytics Dashboard

7 Customer Intelligence Analytics

Several business intelligence dashboards have been implemented.

7.1 KPI Dashboard

The KPI dashboard displays:

- Total Revenue
- Total Transactions
- Total Units Sold
- Anomaly Cases

Current platform statistics:

- Revenue: TZS 221.7 Billion
- Transactions: 28.8 Million
- Electricity Units Sold: 627.4 Million Units
- Anomaly Cases: 38,068

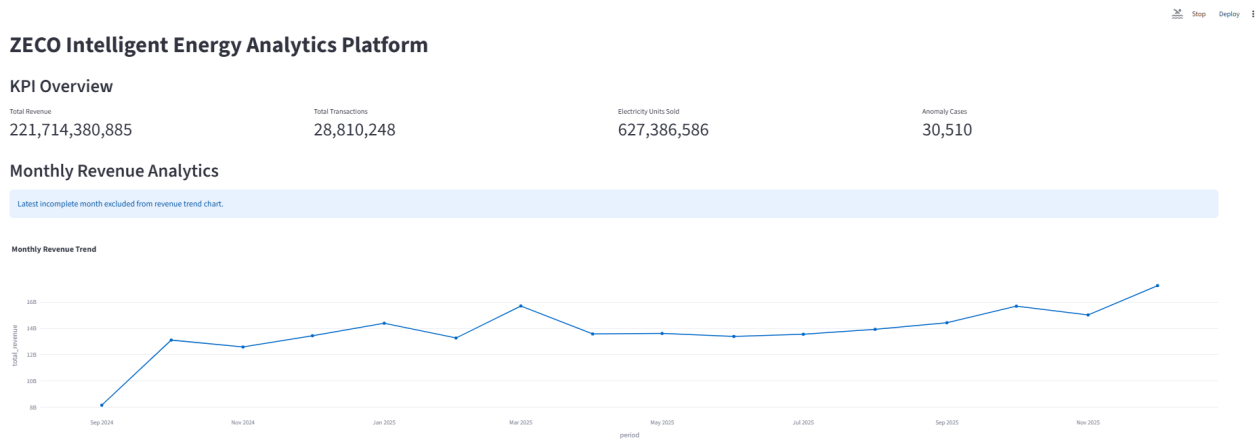


Figure 3: Live Streaming Analytics Dashboard

7.2 Revenue Analytics

Revenue analytics monitors how electricity vending revenue changes over time. Monthly transaction records aggregated using Apache Spark and visualized through interactive dashboards.

The dashboard presents the monthly revenue trend, allowing ZECO management to identify growth patterns, seasonal fluctuations, and periods of increased electricity demand.

The analysis shows a generally increasing revenue trend throughout the observation period, indicating stable growth in prepaid electricity sales. Such information is useful for financial planning, budgeting, and demand forecasting.

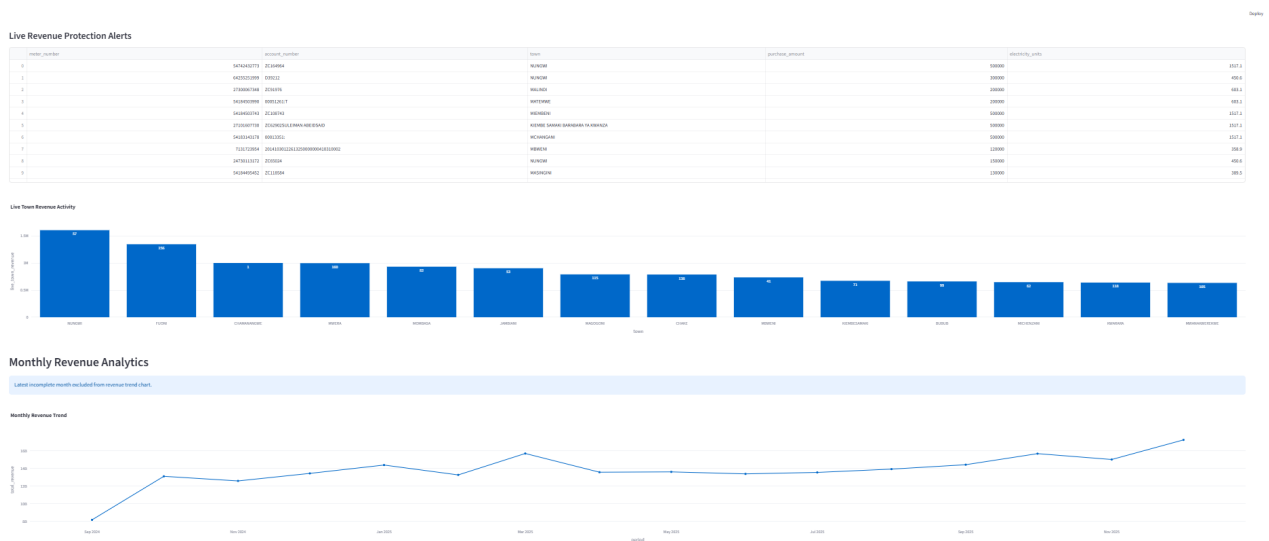


Figure 4: Monthly Revenue Analytics

7.3 Town-Level Analytics

Town-level analytics provide:

- Top revenue-generating towns
- Top electricity-consuming towns
- Monthly payment behaviour

This helps management identify high-value service areas and monitor regional performance.

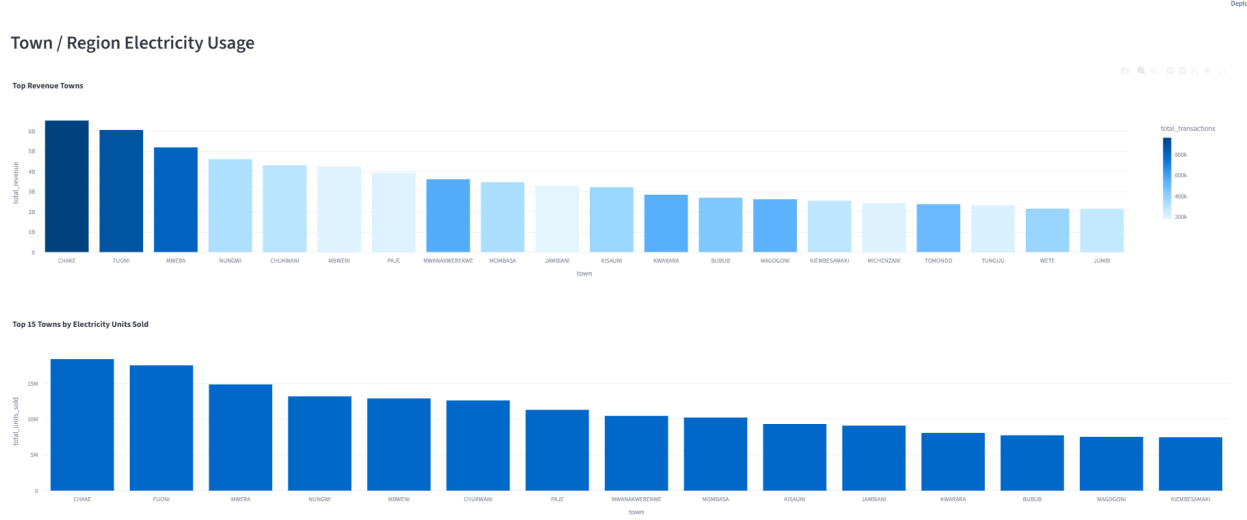


Figure 5: Town Revenue and Consumption Analytics

7.4 Customer Spending Analytics

Customer spending dashboards identify:

- Highest spending customers
- Spending distribution
- Revenue concentration

8 Revenue Protection and Fraud Detection

Revenue protection is a core component of the platform.

The original proposal suggested identifying:

- Irregular vending volumes
- Unusual depletion intervals
- Sudden consumption drops

A hybrid revenue protection framework was implemented by combining Isolation Forest-based anomaly detection with explainable rule-based fraud indicators.

8.1 Fraud Indicators

The implemented fraud indicators include:

1. Irregular Vending Volume
2. Frequent Transactions
3. Unusual Depletion Interval
4. Sudden Consumption Drop
5. Purchase Spike
6. Unit Spike
7. Inactive Customer Behaviour
8. High Variability Behaviour

8.2 Fraud Scenario Score

Each activated fraud indicator contributes to a customer's fraud score.

$$[\text{FraudScenarioScore} = \sum_{k=1}^8 \text{Flag}_k]$$

Higher values indicate stronger evidence of suspicious behaviour.

8.3 Anomaly Classification

Customers are classified into:

- Normal Risk
- Medium Risk
- High Risk

Customer Spending Analysis

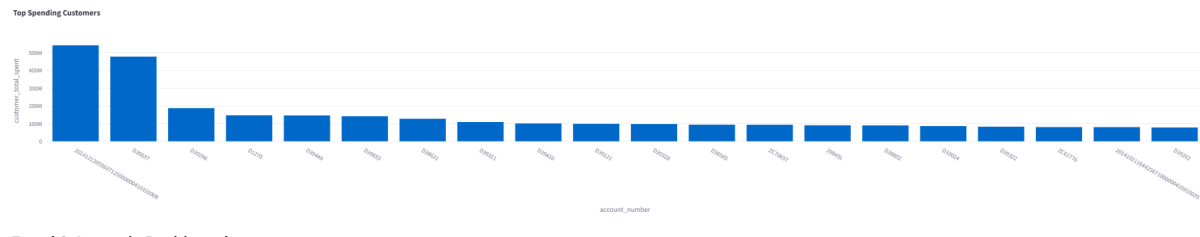


Figure 6: Customer Spending Dashboard

Fraud & Anomaly Dashboard

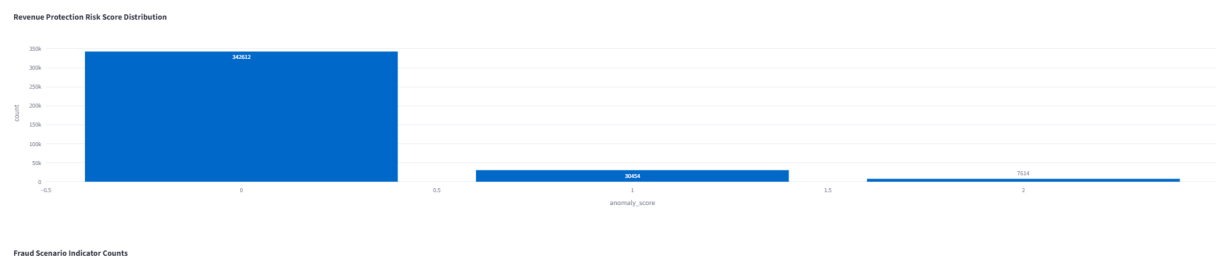


Figure 7: Fraud Anomaly Dashboard

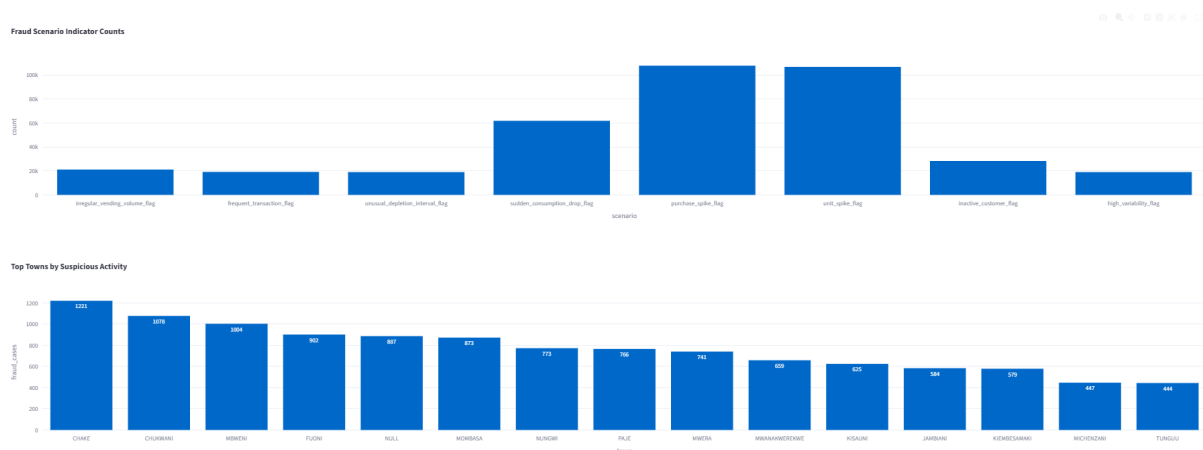


Figure 8: Revenue Protection Dashboard

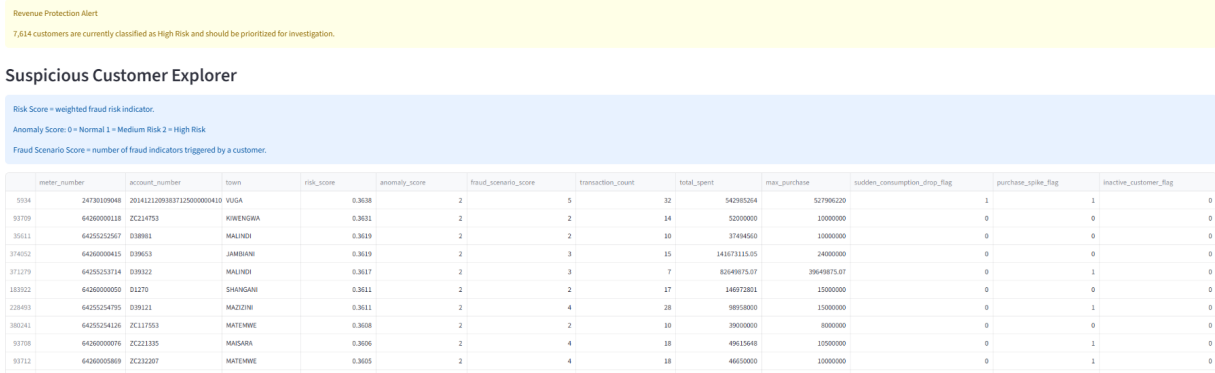


Figure 9: Suspicious Customer Explorer

Current platform results:

- Normal Customers: 342,612
- Medium Risk Customers: 30,454
- High Risk Customers: 7,614

8.4 Revenue Protection Dashboard

The dashboard currently includes:

- Risk Score Distribution
- Fraud Indicator Counts
- Top Towns by Suspicious Activity
- Suspicious Customer Explorer
- Revenue Protection Alerts

This enables ZECO analysts to prioritize investigations and focus on high-risk customers.

9 Forecasting Analytics

9.1 ARIMA Revenue Forecasting

An ARIMA forecasting model has been implemented to predict future revenue.

The model follows: [ARIMA(p,d,q)]

The system currently generates a 30-day revenue forecast.

The forecasting dashboard displays:

- Forecast trend
- Forecast values
- Revenue projections

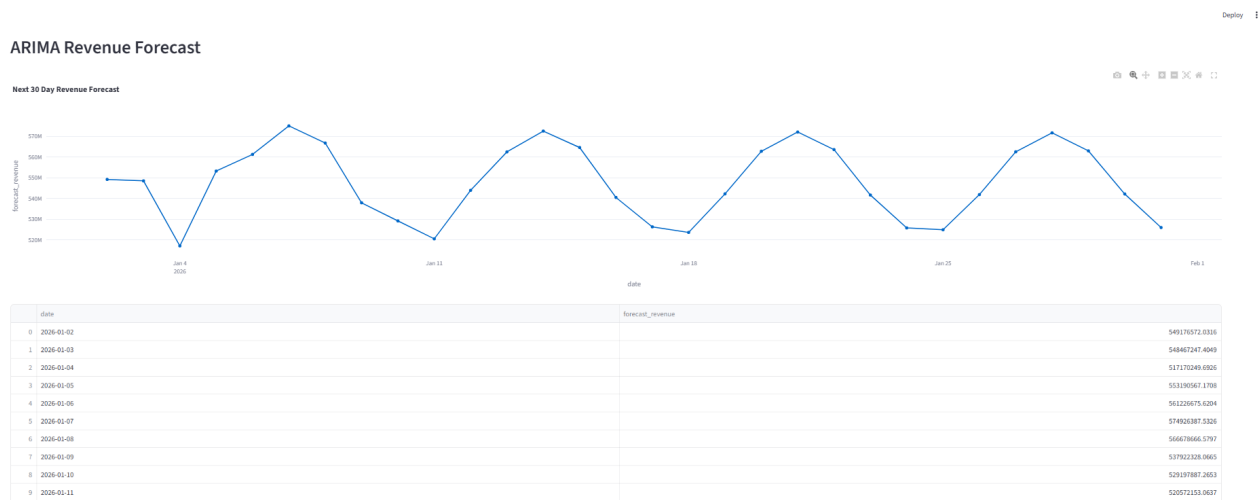


Figure 10: ARIMA Revenue Forecast

9.2 Prophet Token Depletion Forecast

Facebook Prophet was implemented for electricity demand forecasting.

The model is expressed as:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

where:

- $g(t)$ = trend
- $s(t)$ = seasonality
- $h(t)$ = holiday effects
- ϵ_t = random error

The model generates future electricity consumption forecasts and helps support demand planning.

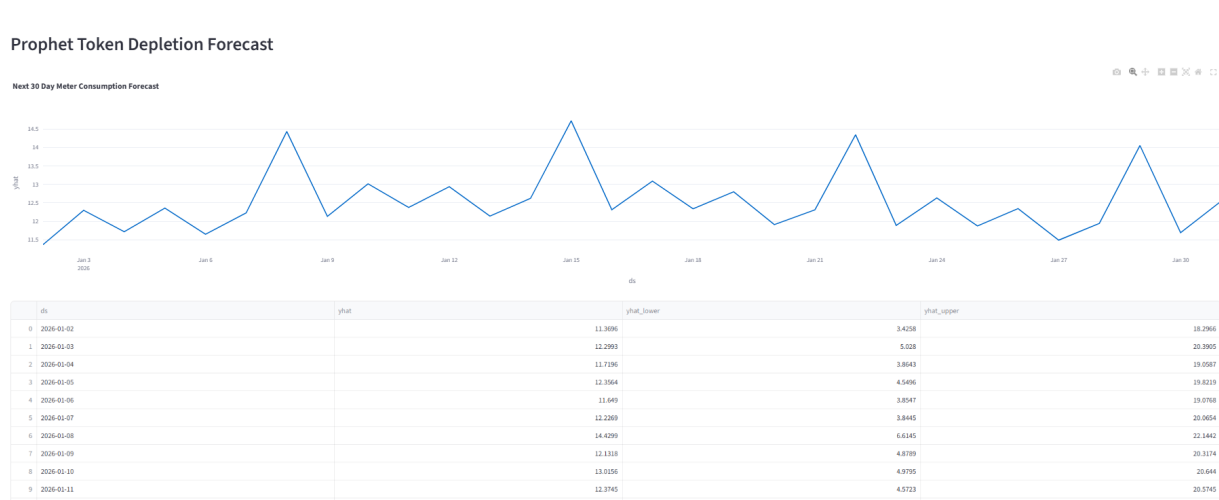


Figure 11: Prophet Electricity Consumption Forecast

10 Current Achievements

The following project components have been completed successfully:

- Bronze-Silver-Gold Lakehouse
- Spark ETL Pipelines
- Customer Intelligence Dashboard
- Revenue Analytics Dashboard
- Fraud Detection Framework
- Suspicious Customer Explorer
- Revenue Protection Alerts
- Spark Structured Streaming
- Live Analytics Dashboard
- ARIMA Forecasting
- Prophet Forecasting
- Town-Level Intelligence Analytics

11 Remaining Work

The following tasks remain:

- Airflow Pipeline Automation
- MLflow Experiment Tracking
- BentoML Model Serving
- Kafka-Based Production Streaming Integration
- Alert Notification System
- Potential Revenue at Risk KPI
- Final System Integration Testing
- Report Finalization

12 Conclusion

The project has successfully progressed from a traditional ETL system into a comprehensive ZECO Revenue Protection and Customer Intelligence Platform.

The implemented system now supports real-time analytics, customer intelligence, fraud detection, forecasting, and business decision support.

Current progress demonstrates that the platform is capable of providing significant operational value for prepaid electricity management while laying the foundation for future machine learning and MLOps enhancements.